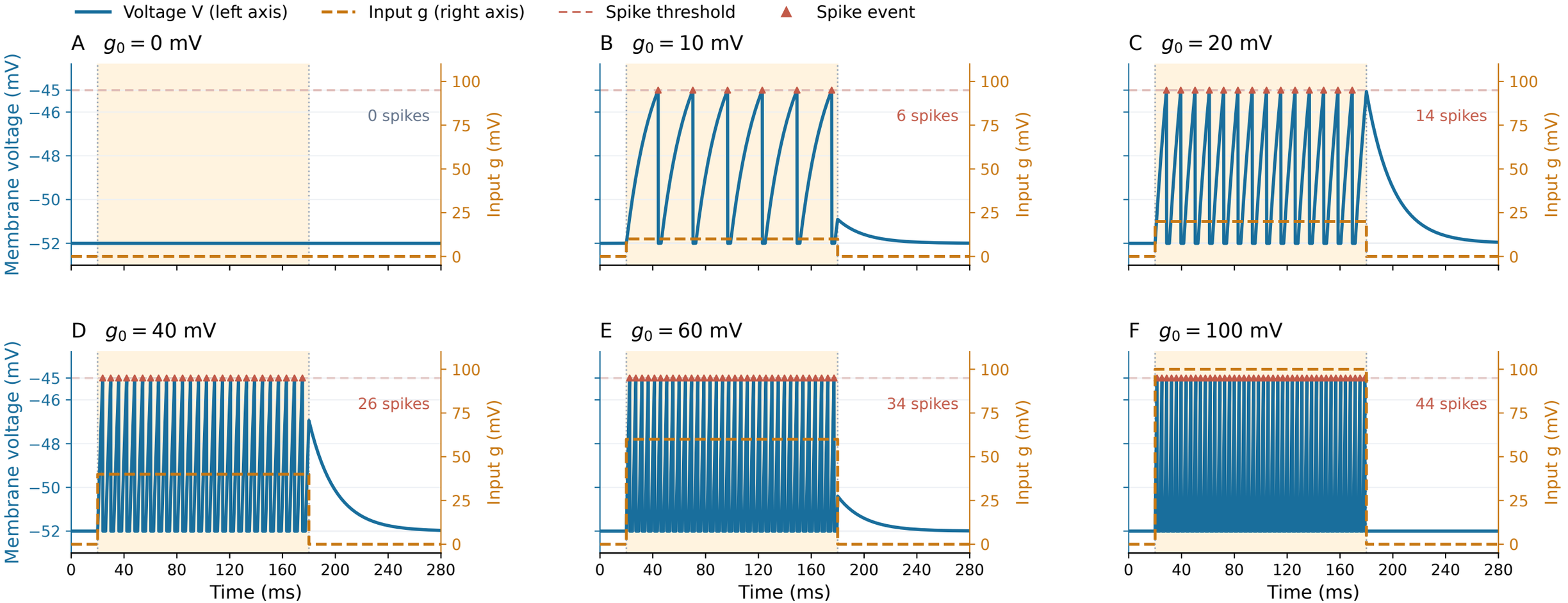
LIF responses to a 160 ms rectangular input pulse

$g(t) = g_0$ from 20 to 180 ms; $g(t) = 0$ otherwise. Input g is independent of spike resets.

$\tau_m \, dV/dt = V_{\text{rest}} - V + g$; $\tau_m = 20$ ms, $V_{\text{rest}} = V_{\text{reset}} = -52$ mV



Orange shading: 160 ms pulse (sbar duration). Input g stays constant throughout the pulse, including during spikes and refractoriness.

Spike threshold: −45 mV; voltage reset: −52 mV; refractory period: 2.2 ms. Action-potential waveforms are not modeled.